Lesson notes · for you

Functions, lists and algorithms: unit check

Programming · Level 3 · Unit check

Time	Materials
20 min	Paper and a pen. No device needed.

Learning goals

- Functions
- Lists and loops
- Searching
- Sorting

Activity: Activity: Break my function

1. Warm-up (3 min): Write one idea from each of the four lessons in your own words.
2. Write (7 min): Write a small function from this unit (largest, find, count_above or is_sorted) with three asserts.
3. Break (7 min): Swap with a partner and write one edge-case assert that you think will break their function. Run it.
4. Fix (3 min): If yours broke, fix it and run every assert again.

Discussion

- Which algorithm surprised you most?
- Which edge case broke the most code today?
- When would you use a set instead of a sorted list?

Check yourself

- One algorithm I can now write and test from memory

Try it this week

Choose one algorithm from this unit (find, binary search, is_sorted or a sort) and write it from memory, with three asserts. Then compare it with the lesson.

Worksheet

Functions, lists and algorithms: unit check

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Name _____ Class _____ Date _____

1 **What does this program print?**

```
def f(x):
    return x * 2

print(f(3) + 1)
```

◯ 7 ◯ 6 ◯ None ◯ An error

2 **A function with no return gives back None.**

◯ True ◯ False

3 **Write count_above(values, limit): it returns how many values are greater than limit.**

```
def count_above(values, limit):
    # return how many values are greater than limit
    return 0
```

Write your program:

4 **total should return the sum of the list, but for [4, 6, 8] it returns 4. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  def total(values):
2      t = 0
3      for v in values:
4          t = t + v
5          return t
```

5 **Match each algorithm to its worst-case work for 1,000 items.**

Linear search	• • • •	about 10,000 comparisons
Binary search	• • • •	about 500,000 comparisons
Bubble sort	• • • •	about 10 looks
Merge sort	• • • •	1,000 looks